

Maharashtra Public Service Commission Assessment

Question Paper Name :	CHROMA1M Biophysics QP2
Subject Name :	Assistant Professor Biophysics Group A Higher and Technical Education Department
Duration :	60
Number of Questions :	100
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes

Assistant Professor Biophysics Group A Higher and Technical Education Department

Group Number :	1
Group Id :	4410092965354
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0

Assistant Professor Biophysics Group A Higher and Technical Education Department

Section Id :	4410092995045
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100
Number of Questions to be attempted :	100
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	4410093006021
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 4410092162612 Question Type : MCQ Option Shuffling : No

A geneticist is analyzing the effects of DNA supercoiling on gene expression in prokaryotic cells. If a mutation leads to a loss of DNA gyrase activity, what is the most likely consequence for the transcription of genes that require open complex formation?

Options :

1. Gene expression will be uniformly enhanced
2. Replication will proceed faster due to easier unwinding
3. Transcription initiation at certain promoters will be inhibited
4. DNA will become more relaxed to promote transcription

Question Number : 2 Question Id : 4410092162609 Question Type : MCQ Option Shuffling : No

A scientist is designing a molecular probe to selectively recognize and bind to specific sequences within the major groove of DNA. Which molecular characteristic of the DNA double helix most directly enables the unique sequence recognition by such probes in the major groove?

Options :

1. The right-handed or left-handed helical nature of the DNA duplex
2. The continuity of the sugar-phosphate backbone surrounding the DNA helix
3. The distinct pattern of hydrogen bond donors and acceptors exposed within the major groove
4. The uniform width of the DNA double helix maintained throughout the molecule

Question Number : 3 Question Id : 4410092162618 Question Type : MCQ Option Shuffling : No

A researcher is investigating the role of disulfide bonds in the quaternary structure of alpha keratin by selectively reducing these bonds in vitro. Predict how this chemical reduction is expected to alter the supramolecular assembly and mechanical characteristics of alpha keratin, considering its hierarchical structure.

Options :

1. Reducing disulfide bonds disrupts interchain crosslinks, leading to decreased fiber rigidity and a loss of the organized supramolecular assembly.
2. Fiber rigidity would increase due to exposure of new hydrophobic regions after reduction.
3. Reduction of these bonds would have no significant effect on the quaternary structure or mechanical strength of alpha keratin.
4. Only hydrogen bonding patterns would be altered, with negligible impact on the overall assembly.

Question Number : 4 Question Id : 4410092162605 Question Type : MCQ Option Shuffling : No

Which characteristic distinguishes ribonucleotides from deoxyribonucleotides at the molecular level?

Options :

1. The presence of a methyl group at the 3' carbon of the pentose sugar
2. The absence of phosphate groups in the nucleotide structure
3. The presence of a hydroxyl group at the 2' carbon of the pentose sugar
4. The presence of an extra amino group on the nitrogenous base

Question Number : 5 Question Id : 4410092162608 Question Type : MCQ Option Shuffling : No

During the process of designing a novel biopolymer for biomedical implants, a team selects a naturally derived polysaccharide due to its low immunogenicity and high biocompatibility. However, they also require the material to retain water and form stable hydrogels under physiological conditions. Which molecular characteristic should the team prioritize to best achieve these functional requirements?

Options :

1. A highly crystalline internal structure with minimal amorphous regions
2. The complete absence of charged residues in the sugar units
3. A high degree of covalent cross-linking with protein molecules
4. The presence of numerous hydrophilic functional groups along the polymer backbone

Question Number : 6 Question Id : 4410092162629 Question Type : MCQ Option Shuffling : No

In a biophysical experiment, a scientist observes that the fractional occupancy of a macromolecule with several identical independent binding sites increases linearly with ligand concentration at low ligand concentrations. Which property of identical independent sites best explains this initial linear relationship?

Options :

1. All binding sites are saturated regardless of ligand concentration
2. Each binding site has the same probability of ligand binding at low occupancy
3. The system reaches binding equilibrium instantaneously at all concentrations
4. Binding at one site changes the affinity of remaining sites

Question Number : 7 Question Id : 4410092162632 Question Type : MCQ Option Shuffling : No

A pharmacologist is studying a protein that contains multiple ligand-binding sites. Binding of a ligand at one site decreases the affinity of a second, spatially distinct binding site for another ligand. Which mechanism best explains this observation?

Options :

1. Positive cooperativity ensures simultaneous binding of multiple ligands.
2. Structural redundancy allows all ligands to bind without interference.
3. Negative allosteric cooperativity reduces ligand binding at the second site.
4. Allosteric activation prevents one ligand from blocking another.

Question Number : 8 Question Id : 4410092162625 Question Type : MCQ Option Shuffling : No

A researcher aims to improve the thermodynamic efficiency of a metabolic pathway by reducing energy dissipation during coupled biochemical reactions. Which strategy would most effectively maximize the fraction of energy conserved for useful cellular work?

Options :

1. Raising the reaction temperature to accelerate metabolic rates and increase overall energy conversion efficiency.
2. Increasing substrate availability throughout the pathway to enhance reaction rates and metabolic energy production.
3. Optimizing enzyme-catalyzed coupling between sequential reactions to minimize energy dissipation and improve efficiency.
4. Designing metabolic pathways that contain only exergonic reactions to maximize the overall energy yield.

Question Number : 9 Question Id : 4410092162628 Question Type : MCQ Option Shuffling : No

If a ligand binds to a macromolecule with positive cooperativity at equilibrium, what is the expected effect on subsequent ligand binding events?

Options :

1. Subsequent ligand binding occurs with increased affinity
2. Subsequent ligand binding occurs with decreased affinity
3. Subsequent ligand binding remains unaffected
4. Subsequent ligand binding is completely inhibited

Question Number : 10 Question Id : 4410092162633 Question Type : MCQ Option Shuffling : No

A researcher observes that the cooperative binding of a substrate to a multimeric enzyme is significantly reduced when a specific small molecule binds at a site distant from the substrate-binding region. Which mechanistic explanation best accounts for how this small molecule modulates the enzyme's behavior through allosterism in this scenario?

Options :

1. The small molecule directly blocks the substrate at the active site region
2. The small molecule induces a conformational change that decreases substrate affinity across all subunits
3. The small molecule alters the catalytic rate but not substrate binding affinity
4. The small molecule stabilizes a state that increases substrate affinity for all subunits

Question Number : 11 Question Id : 4410092162641 Question Type : MCQ Option Shuffling : No

A membrane biophysicist investigates how cholesterol influences the physical properties of a phospholipid bilayer. In an experimental membrane model, cholesterol interacts directly with neighboring phospholipid molecules. Which outcome best describes its effect on membrane fluidity at physiological temperature?

Options :

1. Direct cholesterol-phospholipid contacts result in increased membrane flexibility
2. Direct cholesterol-phospholipid contacts decrease membrane resistance to osmotic pressure
3. Direct cholesterol-phospholipid contacts reduce membrane fluidity by restricting phospholipid movement
4. Direct cholesterol-phospholipid contacts promote local protein clustering in the membrane

Question Number : 12 Question Id : 4410092162644 Question Type : MCQ Option Shuffling : No

A researcher is investigating how the glycocalyx modulates cell-cell interactions within a multicellular tissue model. Considering the roles of the glycocalyx in cellular recognition and adhesion, which scenario best explains how modification of this structure could disrupt communication between adjacent cells without affecting other membrane-associated processes?

Options :

1. Disturbance of lipid microdomains in the cell membrane
2. Reduction of cytoskeletal connections to the membrane glycoproteins
3. Disruption of protein core structures in membrane proteins
4. Alteration of carbohydrate patterns on surface glycoproteins

Question Number : 13 Question Id : 4410092162647 Question Type : MCQ Option Shuffling : No

A research team investigates the impact of fibronectin deficiency on tissue repair in a genetically engineered animal model. They observe delayed wound closure and impaired cell migration. Which mechanistic explanation is most consistent with the established role of fibronectin in mediating cell–extracellular matrix interactions during tissue repair?

Options :

1. Fibronectin regulates the phosphorylation of junctional cadherins in migrating cells
2. Fibronectin facilitates integrin-mediated adhesion and migration of cells through the extracellular matrix
3. Fibronectin binds selectively to proteoglycans for cell-cell signaling in tissue repair
4. Fibronectin provides direct cytoskeletal linkage between adjacent migrating cells

Question Number : 14 Question Id : 4410092162645 Question Type : MCQ Option Shuffling : No

A researcher is investigating how extracellular matrix remodeling directs cell migration during wound healing. Which extracellular matrix component most directly provides structural guidance for directional cell migration during tissue repair?

Options :

1. Hyaluronan increases tissue hydration and regulates swelling responses
2. Elastin confers elasticity and resists deformation during cell migration
3. Laminin networks modulate cell attachment through glycoprotein interactions
4. Collagen fibers align and guide cell movement through integrin-mediated adhesion

Question Number : 15 Question Id : 4410092162648 Question Type : MCQ Option Shuffling : No

A research group explores how mutations in the laminin alpha chain might influence the integrity of the basement membrane during cell-cell interactions. Which scenario best illustrates the effect of such a mutation on tissue architecture and cellular communication within a multicellular environment?

Options :

1. Reduced basement membrane stability disrupts cell polarity and impairs communication
2. Increased deposition of extracellular matrix components without effect
3. Improved adhesion between epithelial cells and connective tissue
4. Enhanced cell migration through increased integrin receptor clustering

Question Number : 16 Question Id : 4410092162804 Question Type : MCQ Option Shuffling : No

Which interaction is commonly measured in atomic force microscopy, a type of scanning probe microscopy?

Options :

1. Van der Waals forces between probe and sample
2. Electric current flow between probe and sample
3. Photon emission from the sample
4. Magnetic resonance from sample spins

Question Number : 17 Question Id : 4410092162799 Question Type : MCQ Option Shuffling : No

Why are lanthanum hexaboride (LaB_6) cathodes preferred over tungsten filaments in certain electron microscopes?

Options :

1. They are less reactive with vacuum contaminants
2. They generate more heat and are easier to fabricate
3. They offer higher brightness and longer lifespan
4. They operate at lower voltages and lower cost

Question Number : 18 Question Id : 4410092162795 Question Type : MCQ Option Shuffling : No

In confocal laser scanning microscopy, what is the primary function of the pinhole aperture in image formation?

Options :

1. It blocks out-of-focus light, improving optical sectioning and image clarity
2. It increases the fluorescence emitted by the specimen
3. It magnifies the image formed by the objective lens
4. It enhances the laser power reaching the specimen

Question Number : 19 Question Id : 4410092162797 Question Type : MCQ Option Shuffling : No

Which statement best explains how wavelength selection impacts the resolving power of an optical microscope in biological imaging?

Options :

1. Using lower intensity light at any wavelength leads to higher resolution.
2. Longer wavelengths improve resolution by increasing contrast between sample regions.
3. Shorter wavelengths improve resolution by reducing the minimum distance that can be distinguished between two points.
4. Increasing numerical aperture alone determines the effect of wavelength on resolution.

Question Number : 20 Question Id : 4410092162803 Question Type : MCQ Option Shuffling : No

A significant source of noise in atomic force microscopy measurements at the nanoscale is thermal fluctuations. At a fixed operating temperature, which modification of the cantilever most effectively minimizes their impact on cantilever performance?

Options :

1. Using a cantilever with a higher spring constant
2. Using a larger tip for imaging
3. Lowering the scan speed
4. Increasing the ambient temperature

Question Number : 21 Question Id : 4410092171764 Question Type : MCQ Option Shuffling : No
A folded dimeric enzyme loses activity when its two subunits separate. Which structural feature has been disrupted?

Options :

1. Disruption of quaternary structure
2. Loss of tertiary structures of each subunit
3. Disruption of α -helix packing within each monomer
4. Alteration of the primary amino-acid sequence

Question Number : 22 Question Id : 4410092171756 Question Type : MCQ Option Shuffling : No
In enzyme kinetics, the initial velocity is measured before appreciable product accumulates. Which of the following influences on the measured forward reaction rate is thereby minimized?

Options :

1. Substrate recognition
2. Active-site formation
3. Reverse reaction influence
4. Enzyme-substrate binding

Question Number : 23 Question Id : 4410092171765 Question Type : MCQ Option Shuffling : No
Which of the following regulatory mechanisms is involved when a high concentration of CTP lowers aspartate transcarbamoylase activity of E. coli without competing with substrates?

Options :

1. Proteolytic enzyme activation
2. Allosteric feedback inhibition
3. Competitive active-site inhibition
4. Covalent enzyme phosphorylation

Question Number : 24 Question Id : 4410092171752 Question Type : MCQ Option Shuffling : No
Which type of inhibition is involved when an inhibitor binds only to the enzyme–substrate complex formed after substrate binding and prevents its productive conversion or product release?

Options :

1. Irreversible inactivation
2. Allosteric activation
3. Competitive inhibition
4. Uncompetitive inhibition

Question Number : 25 Question Id : 4410092171763 Question Type : MCQ Option Shuffling : No
A tetrameric enzyme can contain subunits in different conformational states simultaneously. Ligand binding to one subunit induces a conformational change that alters the affinity of neighbouring subunits sequentially. Which model of allosteric regulation is involved here?

Options :

1. Henderson–Hasselbalch model
2. Koshland–Nemethy–Filmer model
3. Monod–Wyman–Changeux model
4. Briggs–Haldane kinetic model

Question Number : 26 Question Id : 4410092162656 Question Type : MCQ Option Shuffling : No

In an environmental biophysics laboratory, a student observes heat loss from a horizontal plate exposed to still air, where the flow near the surface remains orderly and undisturbed. Which scenario best exemplifies the characteristic behavior of heat conductance in laminar free convection for this situation?

Options :

1. Heat is transferred slowly as layers of air slide smoothly above the plate
2. Heat is transferred instantly by chaotic and random air movement
3. Heat is transferred mostly by strong eddies forming above the surface
4. Heat is transferred rapidly through vigorous mixing and swirling air

Question Number : 27 Question Id : 4410092162654 Question Type : MCQ Option Shuffling : No

A scientist is comparing the rates of molecular diffusion of oxygen in two different aquatic microenvironments: one is a stagnant pond and the other is a fast-moving stream. Both have identical oxygen concentrations at the interface. Based on principles of environmental biophysics, which scenario best explains the expected difference in oxygen diffusion rates between these environments?

Options :

1. The stagnant pond will have a lower molecular diffusion rate due to a thicker boundary layer compared to the fast-moving stream.
2. Oxygen concentration difference alone determines the diffusion rate, regardless of boundary layer characteristics.
3. The fast-moving stream will have a lower molecular diffusion rate due to increased turbulence disrupting the interface.
4. Atmospheric pressure differences are the main reason for changes in molecular diffusion between these environments.

Question Number : 28 Question Id : 4410092162652 Question Type : MCQ Option Shuffling : No

A research team is investigating the transfer of heat energy between the ocean and the atmosphere over a twenty-four hour period. Based on the conservation of energy, which process must be considered when determining the net energy exchange between these two large environmental compartments, especially in the context of diurnal temperature variation and latent heat effects?

Options :

1. The assumption that mechanical mixing is the dominant energy transfer process is correct
2. The exclusion of all radiative transfer processes from calculations is sufficient
3. The inclusion of latent heat transfer during evaporation and condensation events is essential
4. The focus on sensible heat exchange alone fully explains the energy balance

Question Number : 29 Question Id : 4410092162655 Question Type : MCQ Option Shuffling : No

A researcher is comparing the diffusive conductance of the integument in two mammalian species exposed to identical thermal environments. If both species have similar skin surface area and thickness but differ in the composition and organization of their lipid layers, which factor would most likely result in a higher diffusive conductance to water vapor across the integument in one species compared to the other?

Options :

1. Increased ceramide concentration in stratum corneum
2. Higher lipid layer density and more ordered structure
3. Lower lipid layer density and less ordered structure
4. More compact intercellular matrix in the epidermis

Question Number : 30 Question Id : 4410092162661 Question Type : MCQ Option Shuffling : No

In a dense forest ecosystem with a multi-layered plant canopy, how does the heterogeneous arrangement of leaves interact with changing sunlight to influence transpiration throughout the day?

Options :

1. Transpiration varies among canopy layers because differences in light exposure alter stomatal activity and water loss.
2. Transpiration remains nearly identical across canopy layers despite daily changes in light availability.
3. Intermittent cloud cover completely suppresses transpiration across all canopy layers throughout the day.
4. Atmospheric humidity alone governs transpiration uniformly throughout the canopy regardless of light conditions.

Question Number : 31 Question Id : 4410092162666 Question Type : MCQ Option Shuffling : No

A novel drug is hypothesized to enhance recovery following peripheral nerve injury by targeting Schwann cells. Which Schwann cell function is primarily responsible for guiding regenerating axons in the adult peripheral nervous system?

Options :

1. The formation of guidance tubes that direct regrowing axonal sprouts
2. The phagocytosis of myelin debris following axonal damage
3. The regulation of immune cell infiltration at injury sites
4. The secretion of neurotrophic factors into the extracellular space

Question Number : 32 Question Id : 4410092162670 Question Type : MCQ Option Shuffling : No

A patient with altered expression of ion channels in the pacemaker cells is observed to have irregular cardiac rhythms. Considering the genesis of cardiac impulse, which change would most likely impair the normal automaticity of the sinoatrial node without directly affecting the conduction velocity through the atrioventricular node?

Options :

1. Enhanced inward calcium current in ventricular muscle
2. Increased number of gap junctions in atrioventricular node
3. Reduced activity of the pacemaker current channels in sinoatrial node
4. Prolonged opening of potassium channels in Purkinje fibers

Question Number : 33 Question Id : 4410092162677 Question Type : MCQ Option Shuffling : No

During a clinical experiment, a subject develops moderate dehydration after restricted water intake. Laboratory values reveal concentrated urine with low volume. Which renal tubular process is primarily responsible for the observed urinary concentration under these circumstances?

Options :

1. Increased water reabsorption in the collecting duct due to antidiuretic hormone action
2. Decreased aquaporin insertion in the collecting duct reduces water movement
3. Enhanced sodium reabsorption in the proximal tubule independent of hormones
4. Reduced urea reabsorption in the medullary region of the nephron

Question Number : 34 Question Id : 4410092162667 Question Type : MCQ Option Shuffling : No

In a laboratory model, researchers selectively inhibit the production of microtubule-associated proteins in developing neurons without affecting actin filaments or neurofilaments. Which cellular process related to nerve cell morphology and functional connectivity is most likely to be compromised, and how would this manifest in the formation of neural circuits?

Options :

1. Synaptic vesicle release would be inhibited due to decreased calcium influx at presynaptic terminals
2. Dendritic spine formation would be compromised due to altered actin polymerization in the dendrites
3. Axonal outgrowth and guidance would be impaired due to loss of microtubule stability in extending neurites
4. Myelin sheath formation would be delayed due to impaired oligodendrocyte differentiation in the central nervous system

Question Number : 35 Question Id : 4410092162672 Question Type : MCQ Option Shuffling : No

A 60-year-old patient with long-standing hypertension develops concentric left ventricular hypertrophy as an adaptation to chronic pressure overload. Which physiological principle most directly helps maintain stroke volume and cardiac output despite the increased afterload?

Options :

1. Enhanced coronary perfusion improving oxygen delivery
2. Increased sympathetic stimulation increasing heart rate
3. Increased end-diastolic volume leading to enhanced preload
4. Reduced afterload decreasing ventricular workload

Question Number : 36 Question Id : 4410092171772 Question Type : MCQ Option Shuffling : No

In the standard genetic code, which codon signals translation initiation?

Options :

1. AUG
2. UAA
3. UAG
4. UGA

Question Number : 37 Question Id : 4410092171769 Question Type : MCQ Option Shuffling : No

A trait shows high heritability within a particular population studied under relatively uniform environmental conditions. Which interpretation is valid for the observed variation in that population?

Options :

1. Variation is largely genetic in that population
2. Environment has no effect on the trait
3. The trait is fixed across populations
4. The trait cannot change by training

Question Number : 38 Question Id : 4410092171768 Question Type : MCQ Option Shuffling : No

A genetically segregating population shows continuous phenotypic variation, and its population mean changes progressively over successive generations of selection. Which inheritance pattern is most consistent with these observations?

Options :

1. Sex-limited Mendelian inheritance
2. Single-locus codominance
3. Maternal cytoplasmic inheritance
4. Additive polygenic inheritance

Question Number : 39 Question Id : 4410092171773 Question Type : MCQ Option Shuffling : No

A single nucleotide is inserted within the coding region near the beginning of an open reading frame, shifting the grouping of codons downstream from the insertion site. Which property of the genetic code explains this effect?

Options :

1. Wobble-base pairing
2. Degenerate codon usage
3. Non-overlapping triplet code
4. Universal codon meaning

Question Number : 40 Question Id : 4410092171776 Question Type : MCQ Option Shuffling : No

During RecA-mediated recombination, donor ssDNA pairs with an intact duplex. Which condition permits strand exchange?

Options :

1. Nonhomologous DNA ends
2. Terminal inverted repeats
3. Dam-directed methylation
4. Extended sequence homology

Question Number : 41 Question Id : 4410092162686 Question Type : MCQ Option Shuffling : No

A pharmacologist is designing an experiment to test how changes in ionic strength of the solvent impact the function of a reconstituted biological membrane. Which of the following effects is most likely to result directly from increased solvent ionic strength?

Options :

1. Screening of electrostatic charges at membrane surfaces
2. Demixing of lipids within the membrane bilayer
3. Increased spontaneous insertion of membrane proteins
4. Direct alteration of overall membrane thickness

Question Number : 42 Question Id : 4410092162683 Question Type : MCQ Option Shuffling : No

A physiologist investigates how substituting monovalent cations with divalent cations in the extracellular medium impacts the electric double layer structure according to the Gouy-Chapman model. Which outcome is most consistent with this model's treatment of ion valency and the resulting surface potential?

Options :

1. Increased surface charge screening leading to a more compact diffuse layer
2. Reduction in membrane hydrophobicity caused by ion exchange
3. Expansion of the diffuse layer due to lower ion mobility in solution
4. Unchanged double layer structure due to selective permeability

Question Number : 43 Question Id : 4410092162688 Question Type : MCQ Option Shuffling : No

A researcher studying the diffusion of a small uncharged molecule across a biological membrane notices that altering the membrane thickness significantly changes the rate at which the molecule moves. Based on Fick's law, which factor most directly explains this observation in the context of membrane transport?

Options :

1. The temperature maintained around the membrane
2. The structural arrangement of membrane proteins
3. The distance the molecule must traverse through the membrane
4. The charge carried by the molecule during transport

Question Number : 44 Question Id : 4410092162690 Question Type : MCQ Option Shuffling : No

A researcher is investigating the selectivity of a membrane ion channel under physiological conditions. If the channel demonstrates higher permeability to potassium ions than to sodium ions despite similar external concentrations, which structural feature most directly accounts for this ion selectivity?

Options :

1. Selectivity filter residues within the transmembrane protein determine ion-specific transport
2. Lipid microdomains surrounding the channel influence ion selectivity mechanisms
3. Conformational flexibility of the protein channel modulates overall ion throughput
4. Phosphorylation-dependent gating alters the directionality of ion movement

Question Number : 45 Question Id : 4410092162693 Question Type : MCQ Option Shuffling : No

In a genetic mutation study, a patient's skeletal muscle cells display persistent depolarization and impaired relaxation after contraction. Based on the role of ion channels in membrane excitability, which channel dysfunction is most likely responsible for this abnormality and what is the underlying mechanism?

Options :

1. Delayed closing of voltage-gated potassium channels prevents repolarization
2. Impaired voltage-gated calcium channels reduce contraction strength
3. Overactive ligand-gated sodium channels maintain continuous depolarization
4. Non-functional chloride channels destabilize the resting membrane potential

Question Number : 46 Question Id : 4410092162454 Question Type : MCQ Option Shuffling : No

Which term is commonly used to describe the force experienced by particles in a centrifuge during hydrodynamic separation?

Options :

1. Centrifugal force
2. Rotational momentum
3. Sedimentation coefficient
4. Centrifugal acceleration

Question Number : 47 Question Id : 4410092162456 Question Type : MCQ Option Shuffling : No

Which of the following best describes the principle behind analytical ultracentrifugation in hydrodynamic studies?

Options :

1. Particles absorb light differently, allowing their separation during spinning
2. Particles sediment in a centrifugal field based on their mass, shape, and density differences
3. Particles move randomly and separate due to thermal motion only
4. Particles are separated by their electrical charge under a centrifugal force

Question Number : 48 Question Id : 4410092162457 Question Type : MCQ Option Shuffling : No

A researcher is tasked with separating a mixture of proteins using adsorption chromatography. The proteins have similar molecular weights but differ in their surface hydrophobicity and charge. Which strategy would best exploit the basic principles of adsorption chromatography for effective separation in this scenario?

Options :

1. Rely on differences in molecular weight for separation and keep buffer conditions unaltered
2. Select a stationary phase with specific surface chemistry and adjust buffer pH to optimize differential protein binding
3. Use a stationary phase with random surface chemistry and maintain constant buffer conditions
4. Increase the amount of protein loaded to enhance differences in elution profiles

Question Number : 49 Question Id : 4410092162462 Question Type : MCQ Option Shuffling : No

A researcher needs to separate a mixture of proteins with similar molecular weights but different isoelectric points using electrophoresis. Which modification to the electrophoresis system would most effectively achieve this separation?

Options :

1. Increasing the gel concentration to enhance size resolution
2. Applying a higher voltage to increase migration speed
3. Adjusting the pH of the buffer to values between the proteins' isoelectric points
4. Using a non-denaturing buffer to preserve protein conformation

Question Number : 50 Question Id : 4410092162465 Question Type : MCQ Option Shuffling : No

A researcher needs to separate a mixture of proteins with similar sizes but different net charges using an electrophoresis set-up. Which adjustment to the system would most effectively enhance resolution between these proteins?

Options :

1. Using a gel matrix with larger pores for faster movement
2. Altering the pH of the running buffer to change protein charge states
3. Increasing the applied voltage to speed migration
4. Decreasing the sample load to prevent band distortion

Question Number : 51 Question Id : 4410092162470 Question Type : MCQ Option Shuffling : No

Which of the following best describes nanobiomaterials in the context of nanobiophysics?

Options :

1. Materials engineered at the nanoscale for biological applications
2. Materials found only in nature and used for biological purposes
3. Materials used in electronics with no biological function
4. Materials visible to the naked eye used in any industry

Question Number : 52 Question Id : 4410092162481 Question Type : MCQ Option Shuffling : No

Which of the following best explains how surface charge of nanomaterials influences their interaction with biological membranes?

Options :

1. Negatively charged nanomaterials interact more due to mutual repulsion with cell membranes
2. Size of nanomaterials, not charge, is the major determinant in interaction with membranes
3. Neutral nanomaterials interact most strongly because they avoid charge-based interactions
4. Positively charged nanomaterials tend to interact more strongly with cell membranes due to electrostatic attraction

Question Number : 53 Question Id : 4410092162472 Question Type : MCQ Option Shuffling : No

A research team is tasked with producing nanoparticles for biomedical applications using a top-down approach. Which of the following experimental designs best aligns with the principles of a top-down synthesis method?

Options :

1. Utilizing high-energy ball milling to reduce bulk material into nanoscale particles
2. Applying chemical vapor deposition to grow nanoparticles atom by atom
3. Using laser ablation to assemble nanoparticles from a vapor phase
4. Employing biological templates to assemble nanoparticles from solution

Question Number : 54 Question Id : 4410092162477 Question Type : MCQ Option Shuffling : No

During UV-visible spectroscopic analysis of silver nanoparticles, which observation most strongly suggests successful nanoparticle formation?

Options :

1. Emergence of a new absorption band in the visible region
2. Shift in instrument lamp intensity
3. Decrease in baseline noise across the spectrum
4. Drop in background absorption at all wavelengths

Question Number : 55 Question Id : 4410092162479 Question Type : MCQ Option Shuffling : No

Which physical property is primarily measured in Atomic Force Microscopy (AFM) to generate high-resolution images of nanomaterials?

Options :

1. Magnetic field strength at the tip
2. Mass of the cantilever
3. Electron transfer between tip and sample
4. Interatomic forces between probe and sample surface

Question Number : 56 Question Id : 4410092162485 Question Type : MCQ Option Shuffling : No

Which physiological change occurs in the human body at high altitude during flight due to decreased atmospheric pressure?

Options :

1. Reduced oxygen saturation in arterial blood
2. Increased atmospheric carbon dioxide levels
3. Increased partial pressure of oxygen in lungs
4. Increased nitrogen absorption in tissues

Question Number : 57 Question Id : 4410092162484 Question Type : MCQ Option Shuffling : No

A climber rapidly ascending to high altitude experiences symptoms such as headache, nausea, and dizziness. Which physiological mechanism is primarily responsible for these symptoms in acute mountain sickness?

Options :

1. Excessive hemoglobin production in response to hypoxia
2. Increased blood carbon dioxide due to hypoventilation
3. Decreased arterial oxygen saturation due to reduced atmospheric pressure
4. Increased tissue oxygen utilization despite low oxygen

Question Number : 58 Question Id : 4410092162489 Question Type : MCQ Option Shuffling : No

Which factor best explains why astronauts in space experience greater difficulty regulating their body temperature compared to on Earth?

Options :

1. Absence of convection due to microgravity conditions
2. Reduced atmospheric pressure impeding heat loss
3. Increased metabolic heat production in microgravity
4. Exposure to cosmic radiation disrupting cellular function

Question Number : 59 Question Id : 4410092162483 Question Type : MCQ Option Shuffling : No

A researcher is designing a high-altitude acclimatization protocol for astronauts during the first 24 hours of exposure to low atmospheric oxygen pressure. Which physiological adaptation is most critical for preventing hypoxic symptoms?

Options :

1. Enhanced lung ventilation
2. Increased production of red blood cells
3. Decreased tissue metabolism
4. Reduced cardiac output

Question Number : 60 Question Id : 4410092162491 Question Type : MCQ Option Shuffling : No

During long-duration space missions, astronauts are at increased risk for kidney stones. What is the primary physiological reason for this in weightlessness?

Options :

1. Decreased urine volume from dehydration
2. Increased fluid intake in space
3. Increased calcium excretion due to bone demineralization
4. Reduced salt intake during missions

Question Number : 61 Question Id : 4410092171787 Question Type : MCQ Option Shuffling : No

Which DNA sequence feature supports the prediction of a protein-coding gene?

Options :

1. Restriction site cluster
2. Open reading frame
3. Poly(A) tail only
4. Random AT-rich tract

Question Number : 62 Question Id : 4410092171794 Question Type : MCQ Option Shuffling : No

In BLAST output, which result most strongly supports a biologically meaningful sequence match?

Options :

1. Low E-value with high score
2. High E-value with low score
3. Low identity with long gap
4. High gap count with low score

Question Number : 63 Question Id : 4410092171781 Question Type : MCQ Option Shuffling : No

A DNA motif is evaluated against a non-uniform genomic background. Which measure quantifies information gain over that background?

Options :

1. Euclidean distance
2. Relative entropy
3. Hamming distance
4. Pearson correlation

Question Number : 64 Question Id : 4410092171783 Question Type : MCQ Option Shuffling : No

A protein-interaction network keeps function after one paralogous node is lost, but fails after both paralogs are removed. Which conclusion is supported by this observation?

Options :

1. Functional redundancy buffered noise
2. Network entropy became zero
3. Channel capacity was exceeded
4. Mutual information was absent

Question Number : 65 Question Id : 4410092171788 Question Type : MCQ Option Shuffling : No

A nucleotide in a GenBank sequence record is corrected. The stable accession must be retained so that the record remains traceable. Which component is updated?

Options :

1. Accession.version number
2. LOCUS molecule type
3. BioSample organism name
4. KEYWORDS field entry

Question Number : 66 Question Id : 4410092171809 Question Type : MCQ Option Shuffling : No

The Flavr Savr tomato delayed fruit softening by suppressing the expression of the gene encoding which enzyme?

Options :

1. Polygalacturonase
2. ACC oxidase
3. Pectin lyase
4. Cellulase

Question Number : 67 Question Id : 4410092171798 Question Type : MCQ Option Shuffling : No

Which of the following molecular events initiates unidirectional ssDNA transfer during bacterial conjugation?

Options :

1. Primase binding at oriC
2. Transposase cutting IS ends
3. Relaxase nicking at oriT
4. Helicase loading at DnaA

Question Number : 68 Question Id : 4410092171808 Question Type : MCQ Option Shuffling : No

Which of the following plastid features raises recombinant protein yield?

Options :

1. High plastome copy number
2. Maternal plastid inheritance
3. Homologous recombination
4. Polycistronic transcription

Question Number : 69 Question Id : 4410092171805 Question Type : MCQ Option Shuffling : No

For mapping the 5' transcription start site of an RNA by primer extension, where should the gene-specific antisense primer anneal?

Options :

1. Primer binds downstream of start site
2. Primer binds upstream of promoter
3. Primer binds within poly(A) tail
4. Primer binds across splice donor

Question Number : 70 Question Id : 4410092171810 Question Type : MCQ Option Shuffling : No

Which of the following human proteins was first produced in transgenic goat milk?

Options :

1. Human antithrombin
2. Human albumin
3. Human insulin
4. Human interferon

Question Number : 71 Question Id : 4410092162503 Question Type : MCQ Option Shuffling : No

Which device is commonly used to measure the intensity of light in photo-biophysical experiments?

Options :

1. Voltmeter
2. Photometer
3. Thermometer
4. Barometer

Question Number : 72 Question Id : 4410092162501 Question Type : MCQ Option Shuffling : No

Which of the following primarily absorbs the harmful ultraviolet (UV) radiation from the Sun through photochemical filtering?

Options :

1. Atmospheric ozone
2. Leaf pigments
3. Water vapor
4. Soil minerals

Question Number : 73 Question Id : 4410092162500 Question Type : MCQ Option Shuffling : No

Which factor most directly influences the spectral quality of solar radiation reaching the forest floor in a dense tropical rainforest?

Options :

1. Scattering by the atmosphere
2. Transmission through water bodies
3. Reflection from the soil
4. Selective absorption by the canopy

Question Number : 74 Question Id : 4410092162507 Question Type : MCQ Option Shuffling : No

Which cellular process is the primary downstream outcome of photoactivation in the regulation of circadian rhythms?

Options :

1. Protein degradation
2. Signal transduction amplification
3. Membrane potential alteration
4. Gene expression modulation

Question Number : 75 Question Id : 4410092162509 Question Type : MCQ Option Shuffling : No

Which scenario best illustrates how the photodynamic action mechanism can be utilized to disrupt circadian rhythm in cultured mammalian cells?

Options :

1. Exposing cells to UV light and photosensitizer during subjective day, causing general cell damage but not specific circadian disruption
2. Applying a photosensitizer and visible light to cells at their subjective night phase, leading to altered gene expression linked to clock regulation
3. Exposing cells to visible light during subjective day without photosensitizer, resulting in minimal cellular impact
4. Treating cells with photosensitizer in darkness, which does not affect circadian rhythm

Question Number : 76 Question Id : 4410092162821 Question Type : MCQ Option Shuffling : No

In mass spectrometry, why is ion formation critical for sample analysis?

Options :

1. Ionization stabilizes the sample
2. Only ions can be manipulated and detected
3. Ion formation improves sample purity
4. Neutral molecules are preferred for detection but hard to produce.

Question Number : 77 Question Id : 4410092162815 Question Type : MCQ Option Shuffling : No

Why are only paramagnetic substances suitable for ESR analysis?

Options :

1. They are nonpolar and lack magnetic properties.
2. They are always large molecules with complex structures.
3. They have only paired electrons in all orbitals.
4. They contain unpaired electrons that interact with the magnetic field.

Question Number : 78 Question Id : 4410092162818 Question Type : MCQ Option Shuffling : No

In an ESR spectrum, what is the effect of increasing the number of equivalent nuclei of the same nuclear spin interacting with the unpaired electron?

Options :

1. The line width of each peak increases
2. The number of hyperfine lines increases
3. The intensity of the ESR signal increases
4. The separation between hyperfine lines decreases

Question Number : 79 Question Id : 4410092162809 Question Type : MCQ Option Shuffling : No
Which advanced application of fluorescence lifetime measurements can be used to discriminate between different molecular conformations in a complex biological sample?

Options :

1. Steady-state fluorescence spectroscopy
2. Fluorescence correlation spectroscopy
3. Time-resolved fluorescence spectroscopy
4. Synchronous fluorescence spectroscopy

Question Number : 80 Question Id : 4410092162817 Question Type : MCQ Option Shuffling : No
A researcher wishes to analyze the relaxation times (T1 and T2) of a spin system using ESR spectroscopy. Which advanced spectrometer feature is most critical for such time-domain measurements?

Options :

1. Magnet system
2. Amplifier
3. Digital display
4. Pulse generation unit

Question Number : 81 Question Id : 4410092162520 Question Type : MCQ Option Shuffling : No
What does the G-value represent in the context of radiochemistry?

Options :

1. Number of molecules produced or destroyed per 100 eV of absorbed energy
2. Energy absorbed per gram of tissue
3. Concentration of radicals in a sample
4. Rate of free radical decay

Question Number : 82 Question Id : 4410092162526 Question Type : MCQ Option Shuffling : No
Maximum permissible dose (MPD) in the context of radiation exposures can be described as:

Options :

1. the minimum dose required to produce biological effects in human tissues
2. the dose at which all individuals experience radiation sickness
3. the highest dose of radiation that a person may receive without significant risk of adverse health effects
4. the exact amount of radiation received from natural sources annually

Question Number : 83 Question Id : 4410092162516 Question Type : MCQ Option Shuffling : No
A radiology department is planning an experiment to assess the impact of alpha radiation on biological tissues. Which approach would be most appropriate to accurately evaluate alpha ray effects while minimizing other factors related to their interaction with matter?

Options :

1. Use thin tissue sections and minimize air gaps between the alpha source and sample
2. Allow air gaps to reduce exposure and use any tissue thickness
3. Place samples in sealed thick containers to control for external radiation
4. Use thick tissue sections with lead shielding around the source

Question Number : 84 Question Id : 4410092162517 Question Type : MCQ Option Shuffling : No

A researcher observes increased DNA strand breaks in a tissue sample following exposure to ionizing radiation. What would be the reason behind this observation?

Options :

1. Formation of hydroxyl radicals from water radiolysis causing DNA damage
2. Lipid peroxidation resulting in membrane disruption
3. Denaturation of proteins leading to loss of DNA repair enzymes
4. Direct excitation of DNA bases causing chemical modification

Question Number : 85 Question Id : 4410092162525 Question Type : MCQ Option Shuffling : No

A hospital is updating its safety protocols for occupational radiation exposure. Considering the principles of dose equivalent limit (DEL), which approach best aligns with current international guidelines for protecting workers from both natural and man-made radiation sources?

Options :

1. Setting separate limits for natural and man-made exposures without combining them
2. Reporting total exposure annually without adjusting for tissue or radiation type
3. Establishing annual cumulative DELs that account for total effective dose from all occupational exposures, adjusting for radiation type and tissue sensitivity
4. Only considering exposures above background levels for man-made sources

Question Number : 86 Question Id : 4410092162535 Question Type : MCQ Option Shuffling : No

Which of the following is the primary medical use of X-ray imaging?

Options :

1. Measuring electrical activity in muscle tissue
2. Diagnosing blood disorders through cell imaging
3. Visualizing internal bone structures for diagnosis
4. Monitoring brain function in real-time

Question Number : 87 Question Id : 4410092162537 Question Type : MCQ Option Shuffling : No

Which of the following best explains how xeroradiography differs from conventional radiography in medical imaging?

Options :

1. Xeroradiography is used exclusively for dental imaging, while conventional radiography is used for bone fractures.
2. Xeroradiography uses a selenium-coated plate and dry development process, while conventional radiography uses film and wet chemical development.
3. Xeroradiography produces only digital images, while conventional radiography produces analog images.
4. Xeroradiography uses a liquid developer and silver halide, while conventional radiography uses a dry toner.

Question Number : 88 Question Id : 4410092162541 Question Type : MCQ Option Shuffling : No

Which of the following best describes the primary scope of radiotherapy in medical biophysics?

Options :

1. Enhancing the effectiveness of chemotherapy by increasing cellular radiosensitivity
2. Utilization of ionizing radiation for targeted destruction of malignant cells while minimizing damage to normal tissue
3. Providing palliative care to all cancer patients regardless of diagnosis
4. Replacing surgical procedures entirely in cancer treatment

Question Number : 89 Question Id : 4410092162528 Question Type : MCQ Option Shuffling : No

A patient presents with an abnormal electrocardiogram (ECG) showing a prolonged PR interval but a normal QRS complex and QT interval. Based on the principles of electrocardiogram interpretation, which cardiac abnormality is most likely present?

Options :

1. Ventricular fibrillation
2. Second-degree atrioventricular (AV) block
3. Bundle branch block
4. First-degree atrioventricular (AV) block

Question Number : 90 Question Id : 4410092162536 Question Type : MCQ Option Shuffling : No

A hospital is planning to upgrade its mammography unit to enhance early breast cancer detection in patients with dense breast tissue while minimizing radiation dose. Which technological modification is most likely to achieve this goal?

Options :

1. Switching to breast MRI for all screenings
2. Using handheld ultrasound as the primary screening method
3. Implementing digital breast tomosynthesis (DBT) with optimized low-dose protocols
4. Increasing exposure settings in standard 2D mammography

Question Number : 91 Question Id : 4410092171822 Question Type : MCQ Option Shuffling : No

Which technology is widely used for large-scale identification and characterization of proteins in modern proteomics?

Options :

1. Light microscopy
2. Southern blotting
3. Mass spectrometry
4. Karyotype analysis

Question Number : 92 Question Id : 4410092171824 Question Type : MCQ Option Shuffling : No

In a standard bottom-up proteomics workflow, which specific analytical methodology is primarily required to generate fragmented product ions?

Options :

1. Liquid chromatography coupled with tandem mass spectrometry
2. Matrix-assisted laser desorption ionization time-of-flight.
3. Two-dimensional polyacrylamide gel electrophoresis analysis
4. Surface-enhanced laser desorption ionization mass profiling

Question Number : 93 Question Id : 4410092171811 Question Type : MCQ Option Shuffling : No

Which of the following fields maps protein complexes proteome-wide?

Options :

1. Structural proteomics
2. Quantitative proteomics
3. Interaction proteomics
4. Comparative genomics

Question Number : 94 Question Id : 4410092171815 Question Type : MCQ Option Shuffling : No

In the cohesin-based loop extrusion model, which property of CTCF binding sites determines topologically associating domain (TAD) boundary formation?

Options :

1. Convergent orientation of CTCF binding motifs
2. High GC content of CTCF binding motifs
3. Distance of CTCF sites from transcription start sites
4. Presence of CpG island methylation at CTCF sites

Question Number : 95 Question Id : 4410092171820 Question Type : MCQ Option Shuffling : No

Which ratio is commonly used in comparative molecular evolution to test for positive selection acting on protein-coding sequences?

Options :

1. Ratio of nonsynonymous to synonymous substitution rates (dN/dS)
2. Ratio of transition to transversion mutation rates (Ts/Tv)
3. Ratio of intronic to exonic nucleotide content
4. Ratio of GC-rich to AT-rich genomic regions

Question Number : 96 Question Id : 4410092171828 Question Type : MCQ Option Shuffling : No

Which type of neuron is primarily responsible for transmitting information from sensory receptors to the central nervous system?

Options :

1. Motor neuron
2. Purkinje neuron
3. Sensory neuron
4. Interneuron

Question Number : 97 Question Id : 4410092171831 Question Type : MCQ Option Shuffling : No

Which of the following changes makes GluA2-containing AMPA receptor impermeable to Ca^{2+} ?

Options :

1. Palmitoylation of intracellular cysteine residues in the membrane
2. Glycosylation of the extracellular amino-terminal domain during trafficking
3. RNA editing that substitutes arginine for glutamine in the channel pore region
4. Phosphorylation of the C-terminal tail by CaMKII at the synapse

Question Number : 98 Question Id : 4410092171836 Question Type : MCQ Option Shuffling : No

Which cross-frequency interaction normally organizes hippocampal neuronal activity during memory encoding and has been reported to be altered in early amyloid-associated network dysfunction?

Options :

1. Delta-beta phase synchronization
2. Theta-phase-gamma-amplitude coupling
3. Alpha-alpha phase locking
4. Gamma-gamma coherence coupling

Question Number : 99 Question Id : 4410092171827 Question Type : MCQ Option Shuffling : No

Who established the neuron doctrine which states that the nervous system is composed of discrete individual cells rather than a continuous network?

Options :

1. Otto Loewi
2. Santiago Ramón y Cajal
3. Luigi Galvani
4. Alan Hodgkin

Question Number : 100 Question Id : 4410092171837 Question Type : MCQ Option Shuffling : No

Which EEG finding can best distinguish Dementia with Lewy bodies from Alzheimer's disease?

Options :

1. Preserved posterior dominant alpha
2. Prominent posterior slow-wave fluctuations
3. Generalized periodic sharp waves
4. Occipital intermittent rhythmic delta